

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.1: Data Representation

Concept 1.1.4: Two's Complement

Total Questions: 13

Mark Schemes begin on Page 7

Question 2 | Source: 9618_w25_qp_11 (Q1.a.iii)

(iii) Convert the 12-bit two's complement binary integer into denary.

Show your working.

111110111100

Working
.....
.....
.....

Denary value

[2]

Question 3 | Source: 9618_s25_qp_12 (Q2.b.i)

(b) (i) Convert the two's complement binary integer 11100010 into denary.

.....
.....
..... [1]

Question 4 | Source: 9618_s25_qp_12 (Q2.b.ii)

(ii) Write the smallest and the largest two's complement binary integers that can be represented in 8 bits.

Smallest

Largest

[2]

Question 5 | Source: 9618_w24_qp_11 (Q1.b.ii)

(ii) Convert the two's complement binary number into denary.

100110010111

Answer

[1]

Question 6 | Source: 9618_w24_qp_13 (Q8.b)

- (b) Explain how to convert the two’s complement integer 10011111 into denary. Give the denary value after conversion.

Explanation

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.....

.....

.....

Denary value [3]

Question 7 | Source: 9618_w23_qp_12 (Q3.b.i)

- (b) (i) Convert the denary value into a 12-bit two’s complement binary integer.
- 196

Answer [1]

Question 8 | Source: 9618_s23_qp_11 (Q3.d.iv)

- 3 A computer has an Operating System (OS).
- (d) The computer stores data in binary form.
- (iv) Give the smallest **and** largest two’s complement binary number that can be represented using 8 bits.

Smallest

Largest [2]

Question 9 | Source: 9618_s23_qp_12 (Q4.b)

- 4 Data in a computer is stored in binary form.

(b) Give the 8-bit one's complement representation of the denary number -120.

Show your working.

Working

.....

.....

.....

Answer

[2]

Question 10 | Source: 9618_s23_qp_13 (Q7.a)

7 A computer stores data in binary form.

(a) Draw **one** line from each description to its matching denary value.

Description	Denary value
	-127
The smallest integer that can be represented in 8-bit two's complement.	127
	-255
The largest integer that can be represented in 8-bit two's complement.	-128
	-256
The largest unsigned integer that can be represented in 8 bits.	256
	128
	255

[3]

Question 11 | Source: 9618_w22_qp_11 (Q1.a.iii)

(iii) Convert the 8-bit two's complement binary integer into denary.

11100111

Answer [1]

Question 12 | Source: 9618_w22_qp_12 (Q2.a.i)

2 (a) (i) Convert the two's complement binary integer into denary.

10010110

Answer [1]

Question 13 | Source: 9618_w21_qp_12 (Q4.b)

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
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(b) The binary value in the register represents a two's complement binary integer.

Convert the two's complement binary integer into denary.

..... [1]

Question 14 | Source: 9618_s21_qp_11 (Q1.c.iii)

1 Anya scans an image into her computer for a school project.

(c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

.....

.....

.....

.....

Answer (in binary)

[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_13 (Q2.b)

2(b)	1 mark 1111 1001 0100	1
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Mark Scheme for Question 2 | Source: 9618_w25_ms_11 (Q1.a.iii)

1(a)(iii)	1 mark for the working 1 mark for the correct denary value Working: Method 1: 1111 1011 1100 Flip the bits 0000 0100 0011 Add 1 0000 0000 0001 + 0000 0100 0100 Method 2: $-2048 + 1024 + 512 + 256 + 128 + 32 + 16 + 8 + 4$ $// -128 + 32 + 16 + 8 + 4$ Denary value: -68	2
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Mark Scheme for Question 3 | Source: 9618_s25_ms_12 (Q2.b.i)

2(b)(i)	1 mark for: -30	1
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Mark Scheme for Question 4 | Source: 9618_s25_ms_12 (Q2.b.ii)

2(b)(ii)	1 mark for smallest, 1 mark for largest Smallest: 1000 0000 Largest: 0111 1111	2
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Mark Scheme for Question 5 | Source: 9618_w24_ms_11 (Q1.b.ii)

1(b)(ii)	1 mark for: -1641	1
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Mark Scheme for Question 6 | Source: 9618_w24_ms_13 (Q8.b)

8(b)	1 mark for each bullet point for the method (max 2) e.g. • Flip each bit then add 1 ... • ... method of converting the new binary number into denary • Most significant 1 bit is treated as the corresponding negative denary value ... • ... add the other positive corresponding denary values 1 mark for correct conversion Denary value: -97	3
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Mark Scheme for Question 7 | Source: 9618_w23_ms_12 (Q3.b.i)

3(b)(i)	1111 0011 1100	1
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Mark Scheme for Question 8 | Source: 9618_s23_ms_11 (Q3.d.iv)

3(d)(iv)	Smallest: 10000000 Largest: 01111111	2
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Mark Scheme for Question 9 | Source: 9618_s23_ms_12 (Q4.b)

4(b)	1 mark for working; 1 mark for answer • Working: +120 = 0111 1000 • Answer: 1000 0111	2
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Mark Scheme for Question 10 | Source: 9618_s23_ms_13 (Q7.a)

7(a)	1 mark for each correct line	3																							
<table><tr><th>Description</th><th colspan="2">Denary value</th></tr><tr><td rowspan="2">The smallest integer that can be represented in 8-bit two's complement.</td><td>-127</td><td></td></tr><tr><td>127</td><td></td></tr><tr><td rowspan="2">The largest integer that can be represented in 8-bit two's complement.</td><td>-255</td><td></td></tr><tr><td>-128</td><td></td></tr><tr><td rowspan="3">The largest unsigned integer that can be represented in 8 bits.</td><td>-256</td><td></td></tr><tr><td>256</td><td></td></tr><tr><td>128</td><td></td></tr><tr><td></td><td>255</td><td></td></tr></table>			Description	Denary value		The smallest integer that can be represented in 8-bit two's complement.	-127		127		The largest integer that can be represented in 8-bit two's complement.	-255		-128		The largest unsigned integer that can be represented in 8 bits.	-256		256		128			255	
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Mark Scheme for Question 11 | Source: 9618_w22_ms_11 (Q1.a.iii)

1(a)(iii)	-25	1
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Mark Scheme for Question 12 | Source: 9618_w22_ms_12 (Q2.a.i)

2(a)(i)	-106	1
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Mark Scheme for Question 13 | Source: 9618_w21_ms_12 (Q4.b)

4(b)	-51	1
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Mark Scheme for Question 14 | Source: 9618_s21_ms_11 (Q1.c.iii)

1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 $\begin{array}{r} 10 = 0000\ 1010 \\ -10 = 1111\ 0110 \\ \\ 0010\ 0011 \\ + \underline{1111\ 0110} \end{array}$	3
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